Fine-Tuned Open-Weight Language Models for Explainable Energy-Theft Detection: Accuracy, Generalization, and Deployment Cost

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***Abstract*—[EDIT: Draft contains the Experimental Setup and Results sections only; abstract, introduction, related work, and conclusion are written separately.]**

***Index Terms*—Energy theft, non-technical losses, large language models, fine-tuning, smart grid, explainability.**

I. Experimental Setup

Prior work established that a zero-shot proprietary model can outperform supervised classical baselines on fine-grained energy-theft identification [1]. However, a proprietary API is an impractical operating point for a utility provider. Every query-day is billed at API rates, consumption data leaves the utility's infrastructure, and the detector cannot be adapted when new manipulation patterns emerge. This section asks whether small open-weight models, fine-tuned locally, can close that gap, and what such a detector costs, tolerates, and explains. We answer six research questions:

1) **Frontier match (RQ1):** Can fine-tuned open-weight models of at most 14B parameters match the zero-shot proprietary frontier and supervised classical baselines on 13-class theft identification?
2) **Label and compute cost (RQ2):** If open models match the frontier, how many labeled days and how much training compute does it require, and where does performance saturate?
3) **Generalization (RQ3):** If the label bill is affordable, does the detector survive attack types that were absent from its training data?
4) **Severity (RQ4):** Where does any detector fail, as a function of the fraction of energy actually removed?
5) **Explanation (RQ5):** Beyond detection, can the model state the manipulation mechanism and quantify its magnitude, and at what cost to classification?
6) **Deployment (RQ6):** Given the answers above, what does the cost–performance frontier look like against API zero-shot and classical ML, and where is the break-even?

The remainder of this section specifies the shared apparatus: the dataset and its frozen partition, the attack taxonomy, the models and their single shared recipe, the baselines, the metrics, and the statistical protocol. The design logic that connects the experiments is stated last, in Section I-F.

A. Task, Dataset, and Data Partitioning

The task is per-day classification of smart-meter consumption. Given one query day of readings together with the consumer's preceding reference week, the model emits a structured output that declares the day *original* or *manipulated* and, when manipulated, identifies which of twelve attack mechanisms produced it. The model input is text-serialized numeric data. Each sample contains a metadata line (consumer, month, week, and day identifiers), per-day summary statistics (minimum, maximum, mean, and standard deviation), and calendar features (month, day-of-week, day-of-month). It then carries the raw readings themselves: 96 fifteen-minute values for each of the seven reference days and the query day, 768 numbers in total. No features beyond these are added.

All experiments draw from a single dataset of 8,239 labeled consumer-days spanning 324 consumers and all 13 classes. We partition it once, before any experiment, into a consumer-disjoint 80/20 split. The 259 training consumers contribute 6,589 days; the remaining 65 consumers contribute the 1,650 evaluation days. Consumer disjointness matters because per-consumer consumption habits are strong identifiers. A random row-level split would leak consumer identity from train to test and inflate every downstream number. The evaluation slice contains 130 benign days and 1,520 manipulated days (105–130 per attack class), an 11.7:1 imbalance. Every experiment in this paper uses this one partition.

During fine-tuning, each training example pairs the serialized readings with the correct structured answer, which comprises the binary verdict, the attack identification, and the day's reference explanation (Section I-E). The loss is computed on the answer tokens only; the model is never trained to reproduce its $\sim$7,900-token numeric input. At evaluation, the model decodes deterministically and its answer is parsed for scoring.

B. Attack Taxonomy

Table I lists the 13-class taxonomy: the benign class plus twelve synthetic attack mechanisms drawn from the non-technical-loss literature [2], [3]. Each mechanism is a distinct transformation of the daily load profile. The taxonomy spans the practically relevant spectrum, from total meter bypass (attack0) to manipulations that leave total energy unchanged and alter only the profile's shape (attack10). For the generalization

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TABLE I
THE 13-CLASS TAXONOMY: THE BENIGN PROFILE PLUS TWELVE ATTACK MECHANISMS, WITH THE FAMILY GROUPING USED FOR THE OUT-OF-CLASS ROTATIONS OF SECTION II-C.

Class	Mechanism	Effect on profile	Family
original	unmodified consumption	—	—
attack0	all samples set to zero for the whole day	removes trend and magnitude	Z
attack1	all samples × one random factor $\in (0, 1)$	trend kept, magnitude reduced	M
attack2	a random contiguous window zeroed (on–off)	changes trend and amplitude	Z
attack3	each sample × its own random factor $\in (0, 1)$	changes trend and amplitude	M
attack4	readings reduced inside a random sub-period	changes trend and amplitude	M
attack5	each sample ← mean consumption × random $\in (0, 1)$	replaces the pattern	M
attack6	values above a random cut-off clipped to it	changes trend and amplitude	T
attack7	a random cut-off subtracted; negatives to zero	trend kept, total reduced	T
attack9	every sample ← the day's average (flat profile)	removes all trend	S
attack10	the daily pattern reversed in time	trend changed, total unchanged	S
attack12	profile of a lower-consuming user substituted	false data injection	S
attack13	each interval reports the true value or zero	irregular; trend and amplitude	Z

Families: Z = zero-suppression, M = multiplicative/randomized, T = threshold, S = shape substitution.
attack13 is the IEEE sampled-zeroing attack; attacks 8 and 11 of the source taxonomy are not present in the dataset.

TABLE II
THE FROZEN FINE-TUNING RECIPE SHARED BY EVERY RUN IN THIS PAPER. ONLY THE AXIS UNDER STUDY VARIES PER EXPERIMENT: THE BASE MODEL (RQ1), THE TRAINING-SET SIZE (RQ2), THE HELD-OUT ATTACK FAMILY (RQ3), THE OUTPUT TARGET (RQ5), AND THE LORA RANK (APPENDIX A).

Knob	Value
Adapter	LoRA (DoRA), $r = 16$, $\alpha = 32$, dropout 0.05
Target modules	all linear layers
Quantization	bf16 (31B model: 4-bit NF4 QLoRA)
Epochs / batch	1 epoch, batch 1 × 8 gradient-accumulation steps
Learning rate	2×10^{-4}, cosine schedule, 3% warmup
Weight decay	0.01
Max sequence length	8,192–11,008 tokens (tokenizer-dependent)
Decoding at evaluation	greedy, 512 new tokens
Hardware	one NVIDIA H100 80 GB

experiments of Section II-C we group the twelve mechanisms into four families by manipulation type: zero-suppression, multiplicative or randomized reduction, threshold clipping or subtraction, and shape substitution.

C. Models and Fine-Tuning Configuration

We screen seven open-weight instruction-tuned models spanning 4B to 31B parameters and five model families: Qwen3-8B and Qwen3-14B [4], Phi-4-reasoning (14B) [5], Ministral-3-14B [6], Llama-3.1-8B [7], Gemma-4-E4B (4B effective), and Gemma-4-31B [8]. Every model is fine-tuned with the identical recipe of Table II, LoRA [9] in its weight-decomposed variant [10]; only the base model varies. This single-recipe design is a deliberate trade-off. It makes the screening comparison clean at the risk of disadvantaging models whose optimal recipe differs, and we validate the choice retrospectively in Appendix A. The 31B model exceeds the memory budget for 16-bit adaptation on a single GPU and is trained with 4-bit quantized adaptation [11] instead; we flag this asymmetry where it matters.

Two evaluation choices require explanation. First, after training, the low-rank adapter weights are folded back into the base model, so each evaluated model is a single standalone checkpoint, the same artifact a utility would deploy, and the reported inference throughput refers to that artifact. Second, a generative classifier outputs a discrete label rather than a continuous score, which would leave threshold-based metrics such as AUC undefined. We recover a score from the model itself: with the output held fixed up to the classification field, the model's probabilities for the two class words are renormalized into a probability of manipulation. This restores AUC and threshold analysis without altering any generated prediction.

D. Baselines

Two kinds of comparison appear throughout. The first is the *zero-shot proprietary frontier*, Gemini 3.1 Pro, evaluated on the same evaluation slice (0.976 binary balanced accuracy, 0.929 macro-F1). The second is a set of *classical baselines* trained on the same training split and evaluated on the same slice: a random forest, CatBoost, XGBoost, a logistic regression, and an unsupervised local outlier factor, all consuming a compact feature vector computed from the same readings and reference week the LLMs receive (energy and shape ratios of the query day against the reference week, dispersion, near-zero fraction, and quarter-day means). These support paired statistical comparisons. CatBoost and XGBoost carry the full metric set; the random forest is recorded with balanced accuracy, specificity, and identification macro-F1. The logistic regression and the local outlier factor provide binary-detection context only, the former because it was fit as a plain binary detector and the latter because an unsupervised outlier score has no thirteen-class output; both are therefore reported as balanced accuracy alone.

E. Evaluation Metrics and Statistical Protocol

Classification metrics : Binary detection is reported as balanced accuracy, F1 on the manipulated class, AUC-ROC (via verbalizer scores), specificity, precision, and recall. Identification is reported as macro-F1, weighted-F1, balanced accuracy, and per-class F1 over the 13 classes. We emphasize specificity alongside recall because the two error types carry asymmetric operational cost. A false positive is a site inspection dispatched to an honest consumer, and at utility scale the binding constraint is usually the inspection budget rather than the detection rate.

Statistical protocol : All headline metrics carry 95% percentile confidence intervals from a cluster bootstrap [12] that resamples the 65 evaluation consumers with replacement ($B = 10{,}000$, one fixed seed throughout). Days from the same consumer share consumption habits and are therefore not independent; resampling whole consumers preserves this dependence, where a per-day bootstrap would understate the interval width. Paired model comparisons combine the same bootstrap, applied to the metric difference, with McNemar's test on the per-day predictions [13], using the exact binomial form when discordant pairs are fewer than 25. When a family of comparisons is tested together, the family-wise error rate is controlled with Holm's correction [14].

Explanation metrics : Each day in the dataset carries a reference rationale written by a teacher model under the standard self-taught-reasoning protocol [15]. The teacher diagnoses the day from the readings alone, without the ground-truth label; only where its diagnosis is wrong is the label revealed and the rationale regenerated (the *rationalization* fallback, about 11% of days). Rationale quality is measured by ROUGE-L

[16] against this reference and, more stringently, by parameter-extraction fidelity. Five of the twelve mechanisms are governed by a single recorded value: the retained fraction for attack 1, the zeroed window for attack 2, the scaled sub-period for attack 4, and the clipping or subtraction cut-off for attacks 6 and 7. For these we check whether the explanation states the value that actually generated the attack. The remaining seven mechanisms draw an independent random value per reading or replace the profile wholesale, so no single parameter exists to restate. Magnitude estimates (Section II-F) are scored by mean absolute error against the realized energy-reduction fraction, rank correlation, bucket accuracy, and window intersection-over-union.

Severity proxy : Several analyses stratify by theft severity using $\rho = 1 - E_{\text{query}}/\overline{E}_{\text{reference week}}$, the fraction of expected energy removed, with bucket edges at 10%, 30%, and 60%. We measure severity against the reference week rather than against the pre-attack profile of the same day for two reasons. The counterfactual clean day is unobservable in deployment, whereas the reference week is exactly the context the detector receives, so this proxy is computable identically in evaluation and in production. It is also defined for benign days, which the same-day comparison is not, and this yields the calibration that validates the bucket edges: benign day-to-day variation of ρ spans -0.197 to $+0.178$ (5th to 95th percentile), so the 30% edge sits outside natural variation. Attacks 9 and 10 do not change the day's total energy; they only reshape the consumption profile (median $\rho \approx 0$). We call these the *energy-neutral* pair and report them separately rather than forcing them into severity buckets.

F. Experimental Design and Research Questions

A single screening pass (RQ1) fine-tunes all seven models once under the shared recipe. From its results, two reference models are selected by a rule fixed before the screening ran: the *flagship* is the model with the best macro-F1, and the *workhorse* is the smallest model whose paired-bootstrap 95% confidence interval on the macro-F1 gap to the flagship contains zero, with lower inference cost as the tie-break. All further fine-tuning, covering the data-efficiency curve (RQ2), the unseen-attack rotations (RQ3), the rationale and magnitude extensions (RQ5), and the rank validation of Appendix A, is carried out on the workhorse alone. The flagship re-enters in the severity analysis (RQ4) and the deployment frontier (RQ6), both of which re-use the screening predictions. **[EDIT: Experiment-dependency diagram (screening → selection → workhorse experiments → deployment): TikZ or draw.io; the draft ships without it.]**

II. Results and Discussion

In this section, we present and discuss the results. The screening pass answers whether fine-tuned open models reach the zero-shot frontier and selects the workhorse model that all further training runs on. We then cost that result in labels and compute, and stress it against attack families the model never saw. The failure mechanism exposed by that stress test is formalized into a severity analysis, which sharpens the question of what an LLM offers that a much cheaper classical model does not; the rationale-distillation and magnitude experiments answer it. The deployment frontier then assembles accuracy, cost, and capability into an operating recommendation.

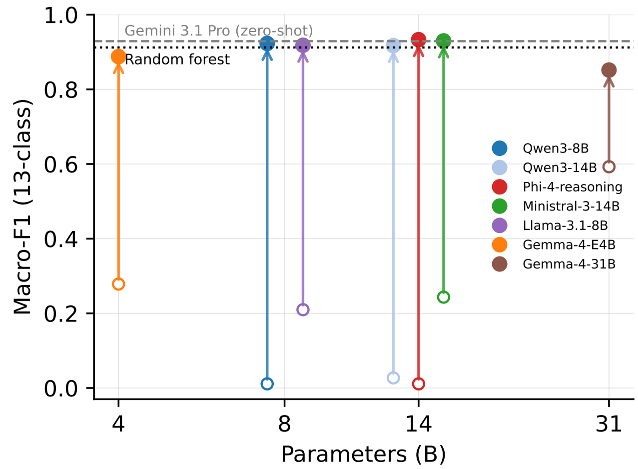

Fig. 1. Macro-F1 against parameter count before and after fine-tuning. Each arrow joins a model's zero-shot score to its fine-tuned score; same-size models are dodged horizontally for legibility. The dashed band is the zero-shot Gemini 3.1 Pro score and the dotted band the random forest baseline.

A. Screening and Model Selection (RQ1)

This experiment establishes the premise of the paper: whether fine-tuned open-weight models at practical scales reach the zero-shot proprietary frontier that motivated this work. Table III reports all seven models in both their zero-shot and fine-tuned conditions on the 1,650-day evaluation slice, against the zero-shot frontier and the classical baselines. Figure 1 visualizes the fine-tuning lift per model.

The answer to RQ1 is affirmative. The best fine-tuned model, Phi-4-reasoning, reaches 0.9331 macro-F1, which exceeds Gemini 3.1 Pro's 0.929 on the same split, and four of the seven models land within 0.016 of it. However, the zero-shot rows show that none of this competence is present before fine-tuning, and the failures are instructive because they are not uniform. Qwen3-8B and Phi-4-reasoning predict *original* for essentially every day (F1 of 0.0000 at a specificity of 1.0). Llama-3.1-8B and Gemma-4-E4B collapse in the opposite direction and declare nearly every day manipulated (specificity of 0.0231 and 0.0154). Notably, zero-shot Qwen3-8B attains an AUC of 0.9066 while its F1 is exactly zero: the base model already ranks manipulated days above benign ones and lacks only a usable decision boundary. This suggests that fine-tuning chiefly teaches the model where to place that boundary, an interpretation the data-efficiency curve of Section II-B will support directly. The single genuine zero-shot competitor is Gemma-4-31B at 0.5927 macro-F1. Yet the same model, fine-tuned, is the *worst* of the seven at 0.8517, receiving the smallest lift from adaptation. This model is also the only one adapted in 4-bit, so its poor showing conflates scale with quantization and we do not read a scale conclusion from it. The converse case carries no such confound: Phi-4-reasoning enters

TABLE III
MAIN RESULTS ON THE 1,650-DAY CONSUMER-DISJOINT EVALUATION SLICE, IN FIVE BLOCKS. ALL ROWS ARE EVALUATED ON THE SAME SLICE; THE CLASSICAL BASELINES ARE TRAINED ON THE SAME TRAINING SPLIT. BEST VALUE PER COLUMN IS BOLDED WITHIN THE FINE-TUNED BLOCK ONLY.

	Model	Binary detection				13-class identification		
		Bal-acc	F1	AUC	Specificity	Macro-F1	Weighted-F1	Bal-acc
Zero-shot proprietary	Gemini 3.1 Pro	0.976	—	—	—	0.929	—	—
Zero-shot open	Qwen3-8B	0.5000	0.0000	0.9066	1.0000	0.0112	—	0.0769
	Qwen3-14B	0.5053	0.0208	0.9193	1.0000	0.0271	—	0.0858
	Phi-4-reasoning	0.5000	0.0000	0.5863	1.0000	0.0112	—	0.0769
	Ministral-3-14B	0.6893	0.5648	0.9473	0.9846	0.2434	—	0.2986
	Llama-3.1-8B	0.5059	0.9543	0.8425	0.0231	0.2098	—	0.2885
	Gemma-4-E4B	0.5041	0.9560	0.8747	0.0154	0.2784	—	0.3842
	Gemma-4-31B	0.9137	0.9767	0.9841	0.8615	0.5927	—	0.6515
Fine-tuned	Qwen3-8B	0.9406	0.9898	0.9939	0.8923	0.9229	0.9247	0.9210
	Qwen3-14B	0.9541	**0.9928**	0.9947	0.9154	0.9173	0.9188	0.9154
	Phi-4-reasoning	0.9502	0.9924	0.9964	0.9077	**0.9331**	**0.9341**	**0.9321**
	Ministral-3-14B	**0.9569**	0.9921	**0.9972**	0.9231	0.9298	0.9307	0.9287
	Llama-3.1-8B	0.9191	0.9895	0.9939	0.8462	0.9181	0.9197	0.9177
	Gemma-4-E4B	0.9456	0.9878	0.9941	0.9077	0.8878	0.8893	0.8850
	Gemma-4-31B	0.9556	0.9800	0.9890	**0.9462**	0.8517	0.8537	0.8503
Supervised ML	Random forest	0.8475	—	—	0.7154	0.9122	—	—
	CatBoost	0.8145	0.9765	0.9828	0.6462	0.9093	—	0.9098
	XGBoost	0.8202	0.9751	0.9811	0.6615	0.9076	—	0.9075
	Logistic regression	0.8669	—	—	—	—	—	—
Unsupervised	LOF	0.8209	—	—	—	—	—	—

Dashes mark metrics not recorded for that model.

TABLE IV
SELECTING THE WORKHORSE: PAIRED CONSUMER-CLUSTER BOOTSTRAP (65 CLUSTERS, $B = 10{,}000$) ON EACH MODEL'S MACRO-F1 GAP TO THE FLAGSHIP. THE WORKHORSE ROW IS BOLDED.

Model	Params (B)	Macro-F1	Δ vs. flagship	95% CI	Tied
Phi-4-reasoning	14	0.9331	—	—	flagship
Ministral-3-14B	14	0.9298	-0.0033	$[-0.0141, +0.0072]$	yes
Qwen3-8B	**8**	**0.9229**	$\mathbf{-0.0102}$	$[\mathbf{-0.0225}, \mathbf{+0.0021}]$	**yes**
Llama-3.1-8B	8	0.9181	-0.0149	$[-0.0284, -0.0022]$	no
Qwen3-14B	14	0.9173	-0.0158	$[-0.0286, -0.0036]$	no
Gemma-4-E4B	4	0.8878	-0.0453	$[-0.0576, -0.0337]$	no
Gemma-4-31B	31	0.8517	-0.0814	$[-0.1000, -0.0631]$	no

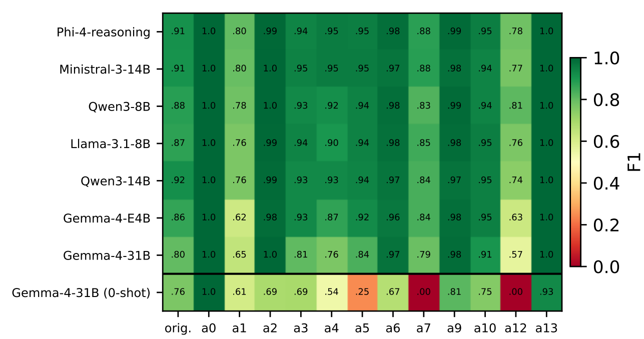

Fig. 2. Per-class F1 for the seven fine-tuned models, with the best zero-shot model (Gemma-4-31B) as the bottom row for contrast. attack1 and attack12 limit every model.

with the weakest zero-shot signal of all (a verbalizer AUC of 0.5863, barely above chance) and leaves as the flagship. In other words, zero-shot competence and fine-tuned competence are different properties, and neither predicts the other in either direction.

Table IV applies the selection rule of Section I-F, fixed before the screening ran: the best model becomes the flagship, and the smallest model statistically tied with it becomes the workhorse. Two models are tied with the flagship, and the smaller of them, Qwen3-8B, is therefore the workhorse. It trails the flagship by 0.0102 macro-F1 with a 95% confidence interval of $[-0.0225, +0.0021]$ that contains zero. One anomaly stands out: Qwen3-14B trails its own 8B sibling by 0.0056 macro-F1 and is *not* tied with the flagship. Within this family, scale did not buy identification quality under the shared recipe. Per-class results (Figure 2; full table in Appendix B) show the same two classes limiting every model. These are attack1, the uniform multiplicative reduction that most resembles a frugal consumer, and attack12, the consumption-profile substitution, with per-class F1 as low as 0.78 and 0.81 for the workhorse against a near-perfect 1.00 on attack0 and attack13. A rank sweep around the shared recipe (Appendix A) finds $r = 8$ statistically tied with the default and $r = 32$ significantly worse, so the screening was not capacity-limited and the single-recipe design was fair.

The classical rows of Table III, however, constrain how these gains should be read. The random forest reaches 0.9122 macro-F1 on the identical split, from the same input, at effectively zero inference cost; CatBoost and XGBoost land

TABLE V
IDENTIFICATION AND DETECTION QUALITY AGAINST TRAINING-SET SIZE FOR THE WORKHORSE AND THE RANDOM FOREST, BOTH TRAINED ON THE IDENTICAL CLASS-STRATIFIED SUBSETS (SEED MEANS OVER THREE DATA SEEDS AT N=65 AND N=260). COST ASSUMES AN H100 AT $2.50/H; THE FOREST TRAINS IN CPU-MINUTES AT EVERY SIZE.

N (per class)	Macro-F1	RF macro-F1	Bin. bal-acc	AUC	Specificity	Train time	Cost
65 (5)	0.2588	**0.7451**	0.4787	0.8869	0.008	3.1 min	$0.13
260 (20)	0.3540	**0.8321**	0.5061	0.9112	0.049	12.5 min	$0.52
1,040 (80)	0.7488	**0.8855**	0.7838	0.9711	0.585	49.7 min	$2.07
6,589 (full)	**0.9229**	0.9122	**0.9406**	**0.9939**	**0.892**	5.23 h	$13.08

Seed-level macro-F1, workhorse, N=65: 0.2427/0.2687/0.2650; N=260: 0.3620/0.3690/0.3310. RF, N=65: 0.7905/0.7327/0.7120; N=260: 0.8473/0.8324/0.8167. CatBoost seed means: 0.7667/0.8624/0.8915; XGBoost: 0.6919/0.8380/0.8790 (per N, ascending). Bold marks the better identifier per row. AUC and specificity at $N \leq 260$ are workhorse seed means.

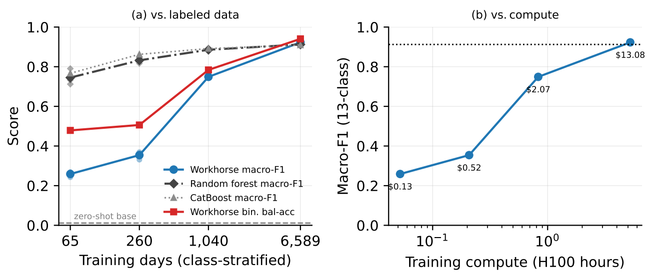

Fig. 3. The data-efficiency curves of the workhorse and the classical baselines trained on the identical subsets: (a) macro-F1 against training-set size, with the workhorse's binary balanced accuracy; (b) workhorse macro-F1 against training compute (the classical models train in CPU-minutes at every size). Per-seed points shown at N=65 and N=260; XGBoost (slightly below the forest throughout) is omitted for legibility and tabulated in Appendix D. Every classical variant dominates until full data.

just below it (0.9093 and 0.9076), so the forest is the classical ceiling here and the tests that follow are against the strongest classical baseline. Paired on the same days, the workhorse's identification advantage over it is not statistically significant ($\Delta = +0.0107$, 95% CI $[-0.0055, +0.0265]$; McNemar $p = 0.204$); only the flagship separates ($\Delta = +0.0209$, CI $[+0.0051, +0.0368]$, $p = 0.0095$), and it does so against all three classical models ($p \leq 0.0037$ for CatBoost and XGBoost). What the workhorse does hold over the forest is binary detection: its specificity of 0.892 against the forest's 0.715 more than halves the false-alarm rate (McNemar $p < 10^{-4}$), and every avoided false positive is an avoided site inspection. Accuracy alone therefore justifies fine-tuning an LLM only weakly, and the following subsections examine what else the model provides. In summary, we can conclude that fine-tuned open models at 8–14B parameters match the zero-shot proprietary frontier. The selection rule hands every subsequent experiment to the workhorse, Qwen3-8B, beginning with the question of what the result costs in labels and compute.

B. Label and Compute Efficiency (RQ2)

Matching the frontier is only useful if the labeling cost is payable, since annotated theft days are precisely what utilities lack. This experiment therefore prices the workhorse's result. We retrain it on stratified subsets of 5, 20, and 80 labeled days per class (three data seeds at the two smallest sizes) and compare against the full 6,589-day run. Table V and Figure 3 report the curve.

Each row of the data-curve table retrains the workhorse with the same recipe on fewer training days, so the rows differ only in the amount of labeled data. Growing the set from 65 to 260 days buys 0.095 macro-F1 (about 0.5 points per 1,000 additional labels at this scale); growing it from 260 to 1,040 buys 0.395 (again about 0.5 per 1,000, sustained over a far larger step); growing it from 1,040 to 6,589 buys only 0.174 (0.03 per 1,000). The marginal value of a label collapses after the third point, which is why we place the elbow between 20 and 80 labels per class: at 1,040 days the model already holds 0.7488, or 81% of the full-data 0.9229. The full run costs 5.23 GPU-hours, roughly $13 at assumed rates, and the 1,040-day point costs $2.07 (per-run costs in Appendix F). The binding budget is label acquisition rather than compute.

The random forest, trained on the identical subsets, supplies the classical comparison at every size, and below full data the comparison is one-sided. The forest holds 0.7451 macro-F1 at 65 days, 0.8321 at 260, and 0.8855 at 1,040, against the workhorse's 0.2588, 0.3540, and 0.7488. It also reaches a usable decision boundary immediately (specificity 0.52–0.77 at $N \leq 260$, where the workhorse sits near zero, so the workhorse at these sizes would flag nearly every consumer for inspection). The conclusion is not an artifact of the estimator: CatBoost tracks slightly above the forest below full data (0.7667/0.8624/0.8915 seed means) and XGBoost slightly below it, so every classical variant we tested dominates the workhorse until labels are plentiful. Only at 6,589 days does the workhorse overtake them, and even there the identification gap to the strongest classical model is not statistically significant, as Section II-A established; the full-data edge that survives testing is binary specificity. In other words, a utility with few labeled days is better served by the forest, and the case for fine-tuning rests on the full-data regime together with the capabilities the rest of this section establishes. However, the more mechanistically interesting finding is in the binary columns. At $N \leq 260$ the model is a near-constant *manipulated* predictor (specificity 0.008–0.049) even though its AUC has already reached 0.887–0.911. The ranking is learned almost immediately; the decision boundary is what takes thousands of examples. This mirrors the zero-shot observation of Section II-A from the opposite direction. It also suggests a practical shortcut we did not pursue: a utility with few labels could combine the early-learned ranking with a threshold calibrated on a small benign sample instead of learning the full boundary. In other words, most of the labeled data buys calibration of a ranking the model forms early.

Identification follows the same order, and earlier than expected. At 65 training days the model already names the correct attack on 35.5% of manipulated days (more than four times the 8.3% chance rate) while its detector is still a near-constant; at 260 days the rate reaches 44.7%. The model learns to name attacks before it learns to withhold a flag on benign days, presumably because the benign class is one of thirteen and calibrating against it requires benign contrast the small subsets barely contain. Notably, the output format is learned before either: every prediction at every training size parses correctly, with zero format violations even at 65 days. This ordering has a deployment reading. The capability the cascade of Section II-G buys from the LLM, adjudicating and naming days a cheap screen has already flagged, is precisely the one

TABLE VI
OUT-OF-CLASS BINARY DETECTION PER HELD-OUT FAMILY ROTATION, EVALUATED ON THE HELD-OUT ATTACKS PLUS ALL 130 BENIGN DAYS, FOR THE WORKHORSE AND THE RANDOM FOREST RETRAINED ON THE SAME REDUCED SETS. HELD-OUT ATTACKS: R-Z = 0, 2, 13; R-M = 1, 3, 4, 5; R-T = 6, 7; R-S = 9, 10, 12.

Rot.	Detector	n	Bal-acc	F1	AUC	Spec.	Recall
R-Z	Workhorse	520	0.9179	0.9562	1.0000	0.8846	0.9513
	Random forest	520	0.8603	0.9546	0.9918	0.7231	0.9974
R-M	Workhorse	635	0.8874	0.9137	0.9435	0.9154	0.8594
	Random forest	635	0.8529	0.8761	0.9192	0.9077	0.7980
R-T	Workhorse	365	0.9325	0.9630	0.9910	0.8692	0.9957
	Random forest	365	0.7973	0.8584	0.9203	0.7308	0.8638
R-S	Workhorse	520	0.8462	0.8497	0.9171	0.9385	0.7538
	Random forest	520	0.8346	0.9312	0.9379	0.7154	0.9538

that emerges first, so even a label-poor utility could pair a classical screen with a lightly trained adjudicator. In other words, labeled data buys the output format first, the attack taxonomy second, and the decision threshold last. In summary, we can conclude that the low-label regime belongs to the classical model, and that the workhorse's frontier match is purchased with the full 6,589 labeled days and $13 of compute. However, every day evaluated so far exhibits an attack each detector saw in training, an assumption no deployment can rely on, and the next experiment removes it, for both of them.

C. Generalization to Unseen Attack Families (RQ3)

Having established that the recipe is affordable, we now remove its most optimistic assumption: that the attack catalog at training time is complete. Real manipulation strategies evolve, and a deployed detector will eventually face a manipulation it was never trained on. We simulate this with four rotations. In each rotation, one entire attack family of Table I is removed from the training data: zero-suppression (R-Z), multiplicative (R-M), threshold (R-T), or shape substitution (R-S). The workhorse is retrained on the remaining attacks and then evaluated only on the held-out attacks plus all 130 benign days, so every manipulated day it sees at evaluation belongs to a family it never saw a training example of. The textual descriptions of the held-out attacks remain in the prompt, so the rotations also measure whether a described-but-never-demonstrated attack can be recognized from its description alone. Crucially, the random forest is retrained on the identical reduced training sets and evaluated on the identical days, so both detector families face the same question and can be compared cell by cell. Table VI reports the per-rotation aggregates for both, Table VII the workhorse's recall on each never-seen attack, and Figure 4 the same per-attack view graphically. **[EDIT: Suggested figure: a small train/eval matrix diagram of the rotation design (four rows, shaded held-out cells) would make the setup graspable at a glance.]**

For the workhorse, detection transfers well. Ten of the twelve never-seen attack types are detected at a recall of 0.85 or higher, five of them at 1.000, and specificity on the benign days is preserved in every rotation (0.869–0.938), so the transfer is not bought with false alarms. The two failures are attack4, a low-severity reduction confined to a sub-window

TABLE VII
DETECTION RECALL ON EACH NEVER-SEEN ATTACK TYPE, WITH THE MEAN REALIZED ENERGY-REMOVAL PROXY $\bar{\rho}$. THE TWO FAILURES, MARKED *, ARE BOTH STATISTICALLY QUIET ATTACKS.

Rotation	Attack	n	Recall	$\bar{\rho}$
R-Z	attack0	130	1.000	1.000
	attack2	130	1.000	0.136
	attack13	130	0.854	0.501
R-M	attack1	126	0.968	0.521
	attack3	130	1.000	0.554
	attack4*	119	0.437	0.084
	attack5	130	1.000	0.500
R-T	attack6	130	0.992	0.372
	attack7	105	1.000	0.649
R-S	attack9	130	1.000	0.043
	attack10*	130	0.269	0.043
	attack12	130	0.992	0.618

*Statistically quiet: $\bar{\rho} < 0.09$ (attack10 is energy-neutral by construction). attack9 is comparably quiet yet fully detected; its flat profile is distinctive by shape.

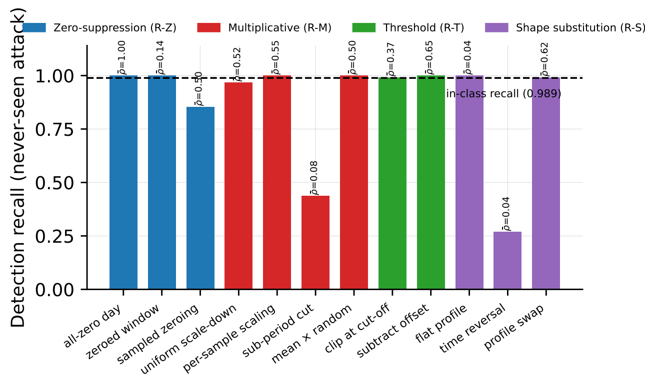

Fig. 4. Detection recall per never-seen attack, grouped by held-out rotation and annotated with the mean energy-removal proxy $\bar{\rho}$. The horizontal line is the in-class recall of the workhorse (0.989).

($\bar{\rho} = 0.084$, recall 0.437), and attack10, the energy-neutral time reversal ($\bar{\rho} = 0.043$, recall 0.269). These two attacks come from different families, one multiplicative and one shape substitution, but they share a statistical property: both barely move the day's energy. Quietness alone does not force a miss, since attack9 is comparably quiet ($\bar{\rho} = 0.043$) yet detected at 1.000; its flattened profile is the most distinctive shape in the taxonomy. This suggests that the ability that carries over to unseen attacks is sensitivity to removed energy, supplemented by shape only where the shape is unmistakable, and that the magnitude of the theft, far more than its mechanism, sets the detection floor.

Identification behaves very differently. On the held-out attacks, the workhorse names the correct attack essentially never: multiclass accuracy is 0.000 in every rotation except R-Z, where attack2 is partially recovered from its prompt description (precision 1.000, recall 0.292). In other words, the model flags the day as manipulated but attributes it to one of the attacks it was trained on, and the description in the prompt rescues attribution only for the single most distinctive mechanism. Naming an attack, unlike detecting it, evidently requires training demonstrations. For a utility this asymmetry is acceptable, since the operational action (inspect) follows from detection, while identification informs the follow-up investigation.

The random forest, facing the same rotations, fails in different places. At the aggregate level, the forest also transfers detection (recall 0.798–0.997). Its specificity sits at 0.715–0.731 on three of the four rotations, essentially its full-catalog level of 0.715, so its false-alarm rate is high with or without a complete catalog; the workhorse holds 0.869–0.938 in the same cells. At the per-attack level, the two detectors fail in different places. On attack4, both fail and the forest fails harder (recall 0.227 against the workhorse's 0.437): where the energy signal is nearly absent, neither representation compensates. On attack10, the time reversal, the forest reaches 0.862 where the workhorse managed only 0.269, because its input features include the shape correlation that the reversal disturbs. On attack6, the threshold clipping, the roles invert: the workhorse holds 0.992 against the forest's 0.754. This suggests that the detection floor has two components, a shared one set by how little energy the attack moves, and a detector-specific one set by what each model's representation happens to encode. The pattern is robust across classical estimators: CatBoost and XGBoost reproduce all three signatures (attack4 at 0.286 and 0.235, attack10 at 0.946 and 0.846, attack6 at 0.715 for both; per-rotation aggregates in Appendix D). The two detector families therefore cover each other's blind spots, which is precisely the property a cascade can exploit.

Worth noting as well is a regularity in the rotation table. The workhorse's specificity barely moves as the training catalog changes (0.869–0.938 across the four rotations, against 0.892 with the full catalog), so its false-alarm rate is evidently anchored in its model of the benign consumer rather than in the attack catalog. This suggests that catalog updates, including retraining on newly discovered attack types, can be deployed without re-budgeting the inspection workload, since the false-positive side of the detector rests on what does not change. In summary, we can conclude that detection survives novel attack families wherever meaningful energy is removed, for both detector families, and that the two fail on different attacks. The workhorse's two failures point at a single governing variable, which the next experiment measures directly.

D. *Severity-Stratified Failure Analysis (RQ4)*

The out-of-class failures suggested that the amount of removed energy governs detectability. This experiment formalizes that hypothesis by stratifying every model's recall over the severity proxy ρ of Section I-E. Table VIII reports recall per severity bucket for the seven screened models and the magnitude variant of Section II-F, with the energy-neutral pair (attacks 9 and 10, which leave the day's total energy unchanged and alter only the profile) reported separately. Figure 5 plots the curves. The full table including multiclass accuracy and the low-data runs is in Appendix C.

The stratification confirms the hypothesis. Every fine-tuned model reaches a recall of 1.000 in the >60% bucket, and the entire quality gradient between models is compressed into the <10% bucket, where the best models hold 0.971 and Gemma-4-31B drops to 0.768. The energy-neutral pair, invisible to any energy-based criterion, is nevertheless detected at 0.954–0.981. This indicates that the models also exploit profile shape, beyond the energy total. For the utility, the reading is direct.

TABLE VIII
DETECTION RECALL BY THEFT-SEVERITY BUCKET (ρ EDGES AT 10/30/60% OF ENERGY REMOVED) ON THE EVALUATION SLICE. THE SEVERITY GRADIENT IS CONFINED ALMOST ENTIRELY TO THE LIGHTEST BUCKET.

Model	<10% (n=138)	10–30% (n=195)	30–60% (n=529)	>60% (n=398)	Energy-neutral (n=260)
Qwen3-8B	0.957	0.990	0.998	1.000	0.969
Qwen3-14B	0.964	1.000	1.000	1.000	0.977
Phi-4-reasoning	**0.971**	1.000	0.998	1.000	0.977
Ministral-3-14B	**0.971**	1.000	0.998	1.000	0.965
Llama-3.1-8B	0.964	1.000	1.000	1.000	0.973
Gemma-4-E4B	0.884	0.995	1.000	1.000	0.969
Gemma-4-31B	0.768	0.964	0.996	1.000	0.954
Qwen3-8B (magnitude)	0.949	1.000	1.000	1.000	0.981

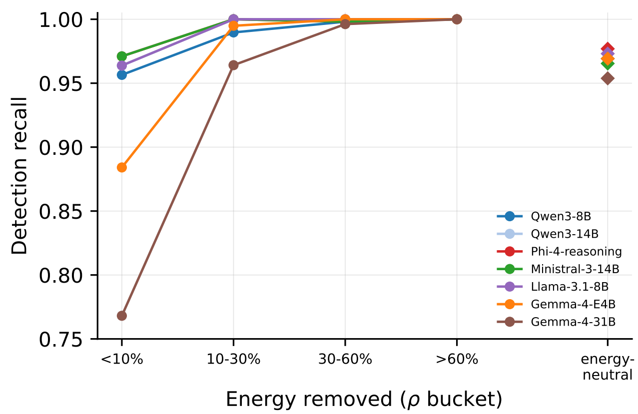

Fig. 5. Detection recall against theft severity for the screened models, with the energy-neutral pair (total energy unchanged, profile reshaped) as detached markers. Every model saturates above 60% energy removal; the differences live below 10%.

Theft large enough to matter financially is detected essentially always, and the residual risk is confined to manipulations that remove less than a tenth of a day's energy, precisely the manipulations whose per-day revenue impact is smallest. However, this finding sharpens rather than settles the objection raised at the end of Section II-A. The random forest also detects consequential theft, as its rotation recalls showed, so if detection saturates for every competent model, the case for an LLM must rest on what it produces beyond the flag. In summary, we can conclude that detectors fail along the severity axis rather than by attack family, and we turn to the capability that classical baselines do not offer at all.

E. *Rationale Distillation (RQ5)*

A flagged day triggers a human decision, and the investigator's first question is *what happened*, since the flag itself already answered *whether*. This experiment asks whether the workhorse can be taught to answer that question, and what the answer costs in classification quality. We fine-tune two variants on the teacher rationales of Section I-E under a catalog-free prompt. The catalog is removed because the screening prompt describes all thirteen mechanisms, and a model could assemble a plausible-sounding explanation by paraphrasing the description of whichever class it predicts; without the catalog, the explanation must be grounded in the readings and in what

TABLE IX
RATIONALE DISTILLATION ON THE WORKHORSE UNDER THE CATALOG-FREE PROMPT: CLASSIFICATION, EXPLANATION OVERLAP (ROUGE-L), AND PARAMETER-EXTRACTION FIDELITY. R0'S MACRO-F1 STEMS FROM A DIFFERENT RECIPE THAN THE SCREENING RUNS AND IS NOT COMPARABLE TO TABLE III.

Run	Train	Macro-F1	Bin. bal-acc	Spec.	ROUGE-L	Param. correct
R0 (label only)	6,589	**0.9444**	0.9384	0.885	—	—
R1 (rationale+label)	6,589	0.9285	0.9412	0.892	**0.454**	**0.798**

Parameter extraction covers the five attacks governed by a single recorded value (Section I-E; n=610).

fine-tuning placed in the weights. R0 is a label-only control; R1 trains on rationale plus label over the full 6,589 days and differs from R0 in nothing else, so any gap between them prices the rationale target itself. Table IX reports classification and explanation quality; the paired contrast, specified before the run, is stated in the text.

Three findings emerge. First, rationale supervision carries a small but real classification cost. Training on rationales costs -0.0159 macro-F1 against the label-only control (95% CI $[-0.0284, -0.0037]$; McNemar $p = 0.0067$ on 93 discordant days), a small but statistically significant identification penalty. Binary detection, however, is statistically unchanged (Δ balanced accuracy $+0.0029$, CI $[-0.0266, +0.0320]$), so the utility pays nothing in detection to gain an explanation for every flag. Second, the explanations are substantive rather than decorative. R1 reaches a ROUGE-L of 0.454 against the teacher and, on the stricter test, states the numerically correct attack parameter on 79.8% of the 610 days from attacks with a stated parameter. The rate tracks how directly the parameter is visible in the readings. The clipping cut-off of attack 6, which appears literally as the day's maximum, is stated correctly on 98.5% of its days; the factor and window parameters of attacks 1, 2, and 4 follow at 82.5–88.5%; the subtracted offset of attack 7, which must be inferred from the residual profile, drops to 37.1%. In other words, the model reliably reports what it can read and struggles with what it must reconstruct. Third, the student's fidelity traces the teacher's own difficulty. R1's overlap with the reference is 0.463 on days the teacher diagnosed unaided but 0.384 on the rationalization-fallback days, where the teacher first erred and was handed the label; the same $\approx$0.46-versus-0.38 split recurs in every full-data run we scored (Appendix E). The fallback marks, by construction, the days whose readings least determine the attack, so the student struggles precisely where the teacher did. This suggests the rationale channel transmits not only the teacher's competence but the difficulty structure of the task itself, and it cautions against reading fidelity as a single number: the hard tail is where explanations degrade first. Two scope limits apply. The planned LLM-judge assessment of rationale quality was not run; a frozen judging package with a pinned rubric is prepared for external scoring. R0's headline macro-F1 of 0.9444, the highest in this paper, comes from a different recipe than the screening runs and therefore does not enter the model comparison of Section II-A. An explanation names the mechanism; an inspector planning recovery also needs its size, which is the final capability we test. In summary, we can conclude that mechanism explanations cost 1.6 macro-F1 points and no detection, and we next extend the same output schema from naming the theft to quantifying it.

TABLE X
EXPLICIT MAGNITUDE QUANTIFICATION BY THE WORKHORSE: HEADLINE ERROR, CALIBRATION, AND LOCALIZATION (TOP), AGAINST PER-DAY CONTROLS AND THE READING-LEVEL CEILING (BOTTOM). MAE IS ON THE ENERGY-REDUCTION FRACTION $\rho \in [0, 1]$, $n = 1{,}260$ MAGNITUDE-BEARING DAYS.

Quantity	Value
MAE (95% CI)	0.0321 $[0.0286, 0.0369]$
Pearson / Spearman	0.985 / 0.976
Severity-bucket accuracy	0.882
Window coverage / mean IoU	0.956 / 0.874
Format-violation rate	0.000

Control	MAE	RMSE	95% CI
Workhorse (LLM)	**0.0321**	**0.0508**	$[0.0286, 0.0369]$
Class-prior (predicted class)	0.0897	0.1378	$[0.0829, 0.0974]$
Class-prior (gold class)	0.0903	0.1385	$[0.0834, 0.0981]$
Ridge on readings	0.1313	0.1602	$[0.1247, 0.1391]$
GBM on readings (ceiling)	0.0012	0.0020	$[0.0011, 0.0013]$

F. Quantifying Theft Magnitude (RQ5)

Extending explanation to quantification, we append a magnitude block to the structured output: the estimated fraction of energy removed and the manipulated time window. The workhorse is fine-tuned to emit it alongside the classification, with ground truth recomputed from the readings via the severity proxy. Two questions matter. The first is whether the estimates are genuine per-day measurements rather than class-conditional averages; the second is whether carrying the extra target degrades classification. Table X answers the first; the interference check, specified before the run, answers the second.

The estimates are genuine. The model attains an MAE of 0.0321 on the energy-reduction fraction with a Spearman correlation of 0.976. It beats the class-prior control (the strategy of looking up the average severity of the predicted attack class) by 0.0576 MAE with a confidence interval excluding zero. The same holds against the prior computed from the gold class, so the advantage is not explained by classification accuracy. Localization is similarly strong (window IoU 0.874, boundary error under one 15-minute interval; per-attack breakdown in Appendix G). The model correctly declines to claim magnitude on the energy-neutral attacks (mean claimed ρ of 0.012 against a true 0.043), and the output format is never violated. However, a gradient-boosted regressor reading the same raw values attains an MAE of 0.0012, roughly 27 times lower. For magnitude in isolation, the readings themselves remain the accuracy ceiling and a dedicated regressor the better tool. The interference check, specified before the run, completes the picture: adding the magnitude target does not degrade classification relative to the plain workhorse (Δ macro-F1 $+0.0057$, CI $[-0.0043, +0.0157]$; McNemar $p = 1.000$ binary, 0.295 multiclass). Set against Section II-E, a pattern emerges: the numerically harder auxiliary task (per-day magnitude regression) costs classification nothing, while the linguistically heavier one (free-text rationales) costs a small but significant 1.6 macro-F1 points. This suggests that the

TABLE XI
THE DEPLOYMENT FRONTIER AT ASSUMED PRICES: IDENTIFICATION QUALITY AGAINST INFERENCE COST PER 1,000 QUERY-DAYS. PARETO-OPTIMAL POINTS ARE BOLDED; THE CASCADE USES THE RANDOM FOREST TO SCREEN AND THE WORKHORSE TO ADJUDICATE FLAGGED DAYS.

Operating point	Kind	Macro-F1	\$/1k days	s/day
Random forest	classical, CPU	**0.9122**	**0.01**	—
Qwen3-8B (workhorse)	local LLM	**0.9229**	**7.48**	10.8
Ministral-3-14B	local LLM	0.9298	9.07	13.1
Phi-4-reasoning (flagship)	local LLM	**0.9331**	**9.07**	13.1
Gemma-4-31B	local LLM	0.8517	33.89	48.8
Gemini 3.1 Pro	API, zero-shot	0.929	32.50	—
Cascade (RF → workhorse)	hybrid	0.9114	6.92	—

Flagship throughput estimated from the same-size Ministral-3-14B; all costs assumption-driven.

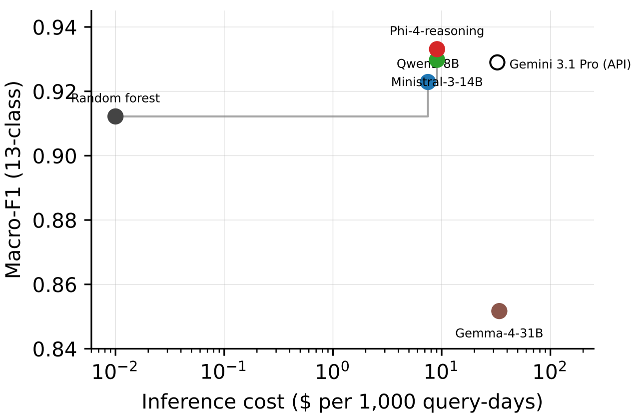

Fig. 6. The cost–performance frontier: macro-F1 against inference cost per 1,000 query-days (log scale). The step line traces the Pareto front; the open marker is the zero-shot API point. The cascade operating point is reported in the frontier table.

price of an auxiliary target tracks the token mass it adds to the training signal rather than the difficulty of the task, since the magnitude block is a handful of tokens and a rationale is a paragraph, and the training losses point the same way (0.014 label-only against 0.42 with rationales). We offer this as an observation from two data points only. In other words, the model's value lies in the integrated output: a single structured answer that reports the detection, the mechanism, the estimated size, and the manipulated window, at no measured cost to the detection it rides on. With the capability set now complete, what remains is to price it. In summary, we can conclude that magnitude estimation is real but ceiling-limited, and the deployment analysis that follows weighs the whole package against its alternatives.

G. Cost–Performance Deployment Frontier (RQ6)

The preceding sections established what the fine-tuned detector achieves, tolerates, and produces. This final experiment assembles those results, at assumed prices, into the deployment decision a utility actually faces. Table XI and Figure 6 place every operating point on a cost–accuracy plane: the random forest on CPU, the fine-tuned models on an amortized local H100 (\$2.50/h), and the zero-shot API reference (\$5 per million tokens, $\sim$6,500 tokens per query-day).

The Pareto front runs random forest → workhorse → flagship, each step trading additional cost for roughly one macro-F1 point. Both the API reference and the 31B model sit strictly inside the front. The API costs 4.3 times the workhorse for 0.006 more macro-F1, and the 31B model costs more than the API while identifying worst of all. Against the API, the local workhorse breaks even after 523 query-days, its one-off \$13.08 fine-tune being repaid at \$0.025 per query-day thereafter. For any utility screening even a modest fleet daily, this is a matter of hours. The frontier prices inference only, so the label bill of Section II-B must be read alongside it: every fine-tuned point on the frontier was bought with the same 6,589 labeled days, and a utility willing to accept 81% of the identification quality can enter the frontier at 1,040 labels and \$2.07 of training. The random forest consumes the identical labeled set, so at full data the frontier comparison is label-neutral. Below full data it is not: Section II-B showed the forest degrading far more gracefully (0.745 macro-F1 at 65 days against the workhorse's 0.259), so the low-label regime belongs to the classical screen outright. The complementary failure profiles of Section II-C are one more reason the cascade pairs the two detectors rather than choosing between them. The question that Section II-A left open, what justifies an LLM against a near-free classical model of equal accuracy, resolves in favor of pairing the two. In the cascade, the near-free random forest screens every day and the workhorse adjudicates only flagged ones, retaining 0.9114 macro-F1 at \$6.92 per thousand days. More to the point, the cascade attaches a mechanism explanation, a magnitude estimate, and a manipulation window to precisely the days a human will review. The LLM rides on infrastructure sized by the screening rate rather than the fleet, so this operating point scales with the inspection workload the utility already budgets for. However, the cascade's flag rate on this theft-heavy evaluation slice (0.925) is far above what a deployed screening population would produce, so the quoted blended cost is an upper bound on the LLM share. In summary, we can conclude that the deployment case for fine-tuned open models rests on what each flag carries: at \$7.48 per thousand days, every flag arrives as an explained and quantified finding rather than a bare score, while the classical screen, which they outperform only narrowly on accuracy, is retained as the free first stage.

H. Threats to Validity and Limitations

While these experiments support the conclusions above, certain limitations should be acknowledged. Firstly, all attacks are synthetic manipulations of one real consumption dataset. The taxonomy is standard in the non-technical-loss literature, but real theft need not respect it, and our own out-of-class results show detection degrading precisely where an attack departs from the energy-removal pattern the taxonomy embodies. One injection artifact deserves a named check: the multiplicative attacks scale readings by floating-point factors, which can leave manipulated values off the meter's native quantization grid, a cue no physical tamper produces. As a planned validation, the workhorse will be re-evaluated on the multiplicative family with the injected values re-quantized to

TABLE XII
LoRA-RANK SWEEP AROUND THE SCREENING DEFAULT ON THE WORKHORSE. PAIRED STATISTICS ARE AGAINST $r = 16$.

Rank	Macro-F1	Bin. bal-acc	AUC	Spec.	Δ vs. r=16	McNemar p
$r = 8$	0.9140	0.9454	0.9943	0.900	-0.0089	0.101
$r = 16$ (default)	**0.9229**	0.9406	0.9939	0.892	—	—
$r = 32$	0.9073	0.9444	0.9930	0.900	-0.0156	0.0158

each meter's step; unchanged recall there would rule the cue out as the source of detection. Secondly, the classical baselines run at library-default hyperparameters. All three tree-family estimators are fully matched across the full-data, data-size, and rotation grids, but dedicated tuning would likely raise their numbers, so we read the classical results as floors; the logistic regression and LOF remain balanced-accuracy only. Thirdly, the rationale evaluation is bounded by its instruments: ROUGE-L rewards overlap rather than truth, parameter fidelity is defined only for the attacks governed by a single recorded value, and the planned LLM-judge assessment was not run. Fourthly, every deployment cost rests on assumed prices (amortized \$2.50/h GPU, blended API tokens). The frontier's ordering is robust to price changes, but its absolute dollar figures are estimates. Finally, the detection floor identified in Sections II-C and II-D, statistically quiet manipulation, is a structural limitation of consumption-based detection shared by every detector we tested. Adversaries aware of it could deliberately operate below the 10% band, at proportionally reduced benefit. Future work will obtain external judge scores from the frozen package, tune the classical baselines, and run the re-quantization check above. None of these threatens the direction of the results, but each would tighten a bound this paper leaves loose.

APPENDIX A
RECIPE VALIDATION: LORA RANK

The screening of Section II-A applied one shared recipe to all seven models, which is only fair if that recipe was not capacity-limiting. We validate this on the workhorse by sweeping the LoRA rank one step below and above the default. Table XII shows that halving the rank is statistically indistinguishable from the default (Δ macro-F1 -0.0089, 95% CI $[-0.0197, +0.0015]$, McNemar $p = 0.101$). Doubling it is significantly *worse* ($\Delta = -0.0156$, CI $[-0.0277, -0.0037]$, $p = 0.0158$). The results suggest that the shared recipe sat at or slightly above the capacity the task requires, so no screened model was starved by it.

APPENDIX B
PER-CLASS RESULTS

Table XIII reports per-class F1 for all seven fine-tuned models with the best zero-shot model as the contrast row; Table XIV the per-class F1 of every zero-shot base; Figure 7 the flagship's confusion matrix. Across every fine-tuned model, attack1 (uniform multiplicative reduction) and attack12 (profile substitution) are the two limiting classes, while attack0 and attack13 are essentially solved.

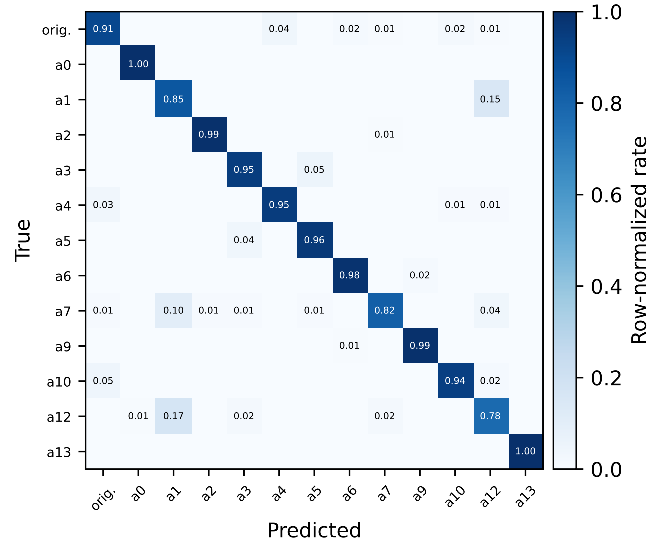

Fig. 7. Row-normalized confusion matrix of the flagship (Phi-4-reasoning) on the evaluation slice. The dominant confusions run between attack1 and the benign class, and between attack12 and the benign class.

APPENDIX C
SEVERITY STRATIFICATION, ALL RUNS

Table XV extends Table VIII to every prediction set in the paper, reporting detection recall and multiclass accuracy per severity bucket. One pattern is worth noting beyond the main text. The low-data runs ($N \leq 260$) show high recall in every bucket only because they flag nearly everything; compare their specificity in Table V. Their multiclass accuracy collapses instead.

APPENDIX D
CLASSICAL GRID, ALL ESTIMATORS

Tables XVI and XVII report the full classical grid (random forest, CatBoost, and XGBoost at library defaults) on the matched data-curve subsets and rotation retrainings. Three observations carry to the main text. First, the ordering below full data is CatBoost > random forest > XGBoost, and every estimator dominates the workhorse until full data. Second, at full data the ordering inverts and the random forest is the classical ceiling (0.9122), which is why the paired tests of Section II-A target it. Thirdly, the rotation signatures (attack4 missed, attack10 caught, attack6 weakened) appear in all three estimators, so the two-component floor of Section II-C describes the whole feature family rather than one model.

APPENDIX E
EXPLANATION QUALITY ACROSS ALL MODELS

Table XVIII reports ROUGE-L, parameter-mention rate, and parameter-correctness for every scored run. Notably, the screening models, trained with the attack catalog in the prompt, produce rationales of the same ROUGE-L band (0.43–0.46) as the dedicated R1 distillation run. This suggests that with catalog-prompt SFT, explanation ability comes essentially free; the catalog-free R1 run matters because it internalizes

TABLE XIII
PER-CLASS F1 ON THE EVALUATION SLICE FOR THE SEVEN FINE-TUNED MODELS, WITH THE BEST ZERO-SHOT MODEL (GEMMA-4-31B) AS THE BOTTOM ROW. BEST VALUE PER CLASS COLUMN IS BOLDED WITHIN THE FINE-TUNED ROWS.

Model	orig.	a0	a1	a2	a3	a4	a5	a6	a7	a9	a10	a12	a13	Macro
Phi-4-reasoning	.911	**.996**	**.805**	.992	.939	**.954**	.951	**.981**	.878	**.989**	**.953**	.783	**1.00**	**.933**
Ministral-3-14B	.909	**.996**	.795	.996	**.945**	.953	**.955**	.969	**.880**	.977	.938	.773	**1.00**	.930
Qwen3-8B	.882	**.996**	.783	**1.00**	.929	.917	.943	**.981**	.826	.988	.941	**.811**	**1.00**	.923
Llama-3.1-8B	.873	**.996**	.763	.992	.941	.899	.943	.977	.852	.985	**.953**	.762	**1.00**	.918
Qwen3-14B	**.915**	**.996**	.765	.988	.929	.928	.940	.969	.836	.973	.946	.739	**1.00**	.917
Gemma-4-E4B	.865	**.996**	.622	.984	.927	.871	.924	.965	.836	.981	.949	.626	.996	.888
Gemma-4-31B	.804	**.996**	.647	.996	.808	.765	.835	.973	.791	.985	.906	.566	**1.00**	.852
Gemma-4-31B (zero-shot)	.762	.996	.613	.689	.694	.539	.248	.670	.000	.815	.750	.000	.929	.593

TABLE XIV
PER-CLASS F1 OF EVERY ZERO-SHOT BASE MODEL ON THE EVALUATION SLICE.

Base model	orig.	a0	a1	a2	a3	a4	a5	a6	a7	a9	a10	a12	a13	Macro
Qwen3-8B	.146	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
Qwen3-14B	.147	.206	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.027
Phi-4-reasoning	.146	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
Ministral-3-14B	.217	.869	.390	.382	.030	.017	.000	.000	.000	.766	.015	.000	.478	.243
Llama-3.1-8B	.040	.743	.152	.405	.000	.180	.292	.124	.000	.733	.015	.000	.043	.210
Gemma-4-E4B	.027	.594	.298	.616	.000	.377	.028	.015	.000	.780	.015	.000	.868	.278
Gemma-4-31B	.762	.996	.613	.689	.694	.539	.248	.670	.000	.815	.750	.000	.929	.593

that ability into the weights. Every run also reproduces the fidelity split of Section II-E: ROUGE-L of 0.44–0.47 on days the teacher diagnosed unaided against 0.34–0.39 on the rationalization-fallback days, so the difficulty signature is a property of the task rather than of one student.

APPENDIX F
TRAINING COST

Table XIX records the training cost of every run, from the per-run metadata. All runs execute on a single H100 80 GB; the amortized figures of Section II-G derive from these wall-clock times at $2.50/h. The 4-bit 31B model is the outlier at 22.4 hours, since quantized training trades memory for time.

APPENDIX G
PER-ATTACK MAGNITUDE ERROR

Table XX decomposes the magnitude MAE of Section II-F per attack. The total bypass (attack0), whose true ρ is identically 1, is estimated exactly. The hardest attacks for magnitude are those whose reduction is confined to sub-windows or applied irregularly (attack4, attack13). On the energy-neutral pair the model claims a mean ρ of 0.012 against a true 0.043; it correctly declines to assert magnitude where there is none to measure. On the 130 benign days it never emits a magnitude block for a day it classifies as original, and its 18 false positives carry a mean claimed ρ of 0.067.

APPENDIX H
REMARK ON THE LEGACY 32B PILOT

An earlier pilot fine-tuned Qwen3-32B with 4-bit QLoRA under the project's original data regime. It was evaluated on a 130-day class-stratified subset of a different holdout, with decision thresholds calibrated on a validation split. Its results (binary balanced accuracy up to 0.977 at a specificity of 1.0 under verbalizer thresholding; multiclass macro-F1 0.833) are not comparable to any table in this paper: different evaluation set, different size, and threshold tuning that the journal-round runs deliberately forgo. We cite them only to acknowledge the pilot's existence. Re-scoring this checkpoint on the present benchmark would make it comparable; we leave that outside the present scope.

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TABLE XV
DETECTION RECALL / MULTICLASS ACCURACY PER SEVERITY BUCKET FOR EVERY PREDICTION SET (EVALUATION SLICE; BUCKET n = 138/195/529/398, ENERGY-NEUTRAL n = 260).

Run	$<10\%$	10–30%	30–60%	$>60\%$	Energy-neutral
Qwen3-8B (workhorse)	0.957 / 0.942	0.990 / 0.923	0.998 / 0.904	1.000 / 0.940	0.969 / 0.950
Qwen3-14B	0.964 / 0.913	1.000 / 0.928	1.000 / 0.898	1.000 / 0.912	0.977 / 0.965
Phi-4-reasoning (flagship)	0.971 / 0.957	1.000 / 0.944	0.998 / 0.923	1.000 / 0.925	0.977 / 0.965
Ministral-3-14B	0.971 / 0.957	1.000 / 0.954	0.998 / 0.906	1.000 / 0.930	0.965 / 0.954
Llama-3.1-8B	0.964 / 0.949	1.000 / 0.949	1.000 / 0.906	1.000 / 0.910	0.973 / 0.962
Gemma-4-E4B	0.884 / 0.848	0.995 / 0.923	1.000 / 0.856	1.000 / 0.872	0.969 / 0.962
Gemma-4-31B	0.768 / 0.761	0.964 / 0.856	0.996 / 0.828	1.000 / 0.837	0.954 / 0.938
Qwen3-8B (magnitude)	0.949 / 0.920	1.000 / 0.949	1.000 / 0.915	1.000 / 0.947	0.981 / 0.962
$r = 8$	0.949 / 0.928	1.000 / 0.918	0.998 / 0.883	1.000 / 0.930	0.977 / 0.954
$r = 32$	0.928 / 0.906	0.990 / 0.923	0.996 / 0.881	1.000 / 0.910	0.989 / 0.958
R0 (label only)	0.957 / 0.949	1.000 / 0.954	1.000 / 0.936	1.000 / 0.957	0.977 / 0.969
R1 (rationale+label)	0.942 / 0.935	1.000 / 0.949	0.996 / 0.913	1.000 / 0.935	0.981 / 0.954
N=1,040	0.928 / 0.804	0.990 / 0.856	1.000 / 0.667	0.995 / 0.754	0.954 / 0.896
N=260 (s42)	0.993 / 0.399	0.985 / 0.523	0.945 / 0.250	0.995 / 0.412	0.996 / 0.946
N=260 (s43)	0.978 / 0.370	0.990 / 0.518	0.896 / 0.268	0.980 / 0.410	0.973 / 0.858
N=260 (s44)	0.993 / 0.478	0.990 / 0.600	0.928 / 0.174	0.982 / 0.417	0.969 / 0.846
N=65 (s42)	0.978 / 0.203	0.969 / 0.380	0.890 / 0.113	0.970 / 0.427	0.989 / 0.800
N=65 (s43)	0.935 / 0.174	0.949 / 0.339	0.964 / 0.187	0.982 / 0.455	0.965 / 0.658
N=65 (s44)	0.935 / 0.138	0.923 / 0.313	0.915 / 0.176	0.965 / 0.455	0.962 / 0.712

TABLE XVI
CLASSICAL DATA-CURVE CELLS: MACRO-F1 PER TRAINING-SET SIZE AND DATA SEED (MATCHED SUBSETS).

N	Seed	Random forest	CatBoost	XGBoost
65	42	0.7905	**0.8126**	0.7180
	43	0.7327	**0.7587**	0.6459
	44	0.7120	**0.7288**	0.7118
260	42	0.8473	**0.8706**	0.8417
	43	0.8324	**0.8675**	0.8245
	44	0.8167	**0.8491**	0.8477
1,040	42	0.8855	**0.8915**	0.8790
6,589 (full)	—	**0.9122**	0.9093	0.9076

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TABLE XVII
CLASSICAL BASELINES UNDER THE ROTATIONS: OUT-OF-CLASS BINARY DETECTION PER HELD-OUT FAMILY (EVALUATED ON HELD-OUT ATTACKS PLUS ALL 130 BENIGN DAYS).

Rot.	Estimator	Bal-acc	Spec.	Recall
R-Z	Random forest	0.8603	0.7231	0.9974
	CatBoost	0.8218	0.6462	0.9974
	XGBoost	0.8385	0.6769	1.0000
R-M	Random forest	0.8529	0.9077	0.7980
	CatBoost	0.8617	0.9154	0.8079
	XGBoost	0.8353	0.8923	0.7782
R-T	Random forest	0.7973	0.7308	0.8638
	CatBoost	0.7520	0.6615	0.8426
	XGBoost	0.7478	0.6615	0.8340
R-S	Random forest	0.8346	0.7154	0.9538
	CatBoost	0.8487	0.7154	0.9821
	XGBoost	0.8038	0.6615	0.9462

TABLE XVIII
EXPLANATION QUALITY FOR EVERY SCORED PREDICTION SET ($n = 1{,}520$ MANIPULATED DAYS; PARAMETER EXTRACTION ON THE FIVE ATTACKS WITH A STATED PARAMETER, $n = 610$).

Run	ROUGE-L	Param. mentioned	Param. correct
R0 (label only)	0.000	0.000	0.000
R1 (rationale+label)	0.454	0.953	0.798
Qwen3-8B (workhorse)	0.455	0.949	0.775
Qwen3-14B	0.456	0.948	0.682
Phi-4-reasoning	0.457	**0.969**	**0.802**
Ministral-3-14B	**0.462**	0.951	0.669
Llama-3.1-8B	0.453	0.949	0.671
Gemma-4-E4B	0.444	0.892	0.712
Gemma-4-31B	0.431	0.890	0.666
Qwen3-8B (magnitude)	0.458	0.966	0.795

TABLE XIX
TRAINING COST PER RUN: TRAINING ROWS, WALL-CLOCK, AND FINAL TRAINING LOSS.

Run	Train rows	Wall-clock (h)	Final loss
Qwen3-8B (workhorse)	6,589	5.23	0.407
Qwen3-14B	6,589	8.29	0.387
Phi-4-reasoning	6,589	5.52	0.424
Ministral-3-14B	6,589	7.52	0.354
Llama-3.1-8B	6,589	4.12	0.410
Gemma-4-E4B	6,589	7.43	0.338
Gemma-4-31B (4-bit)	6,589	22.36	0.373
Qwen3-8B (magnitude)	6,589	5.43	0.371
$r = 8$ / $r = 32$	6,589	5.24 / 5.23	0.406 / 0.408
R-Z / R-M / R-T / R-S	5,035–5,646	3.6–4.5	0.39–0.44
R0 / R1	6,589	4.62 / 4.73	0.014 / 0.424
$N \in \{65, 260, 1{,}040\}$	65–1,040	0.05–0.83	0.56–1.25

R0's low loss (0.014) reflects label-only targets being near-trivially learnable; the gap to R1 reflects the cost of the rationale tokens rather than model quality.

TABLE XX
MAGNITUDE MAE PER ATTACK (MAGNITUDE-BEARING ATTACKS; n PER CLASS AS IN SECTION I-A).

Attack	MAE (ρ)
attack0	0.0000
attack3	0.0272
attack5	0.0279
attack1	0.0291
attack7	0.0300
attack12	0.0300
attack2	0.0419
attack6	0.0426
attack4	0.0454
attack13	0.0473